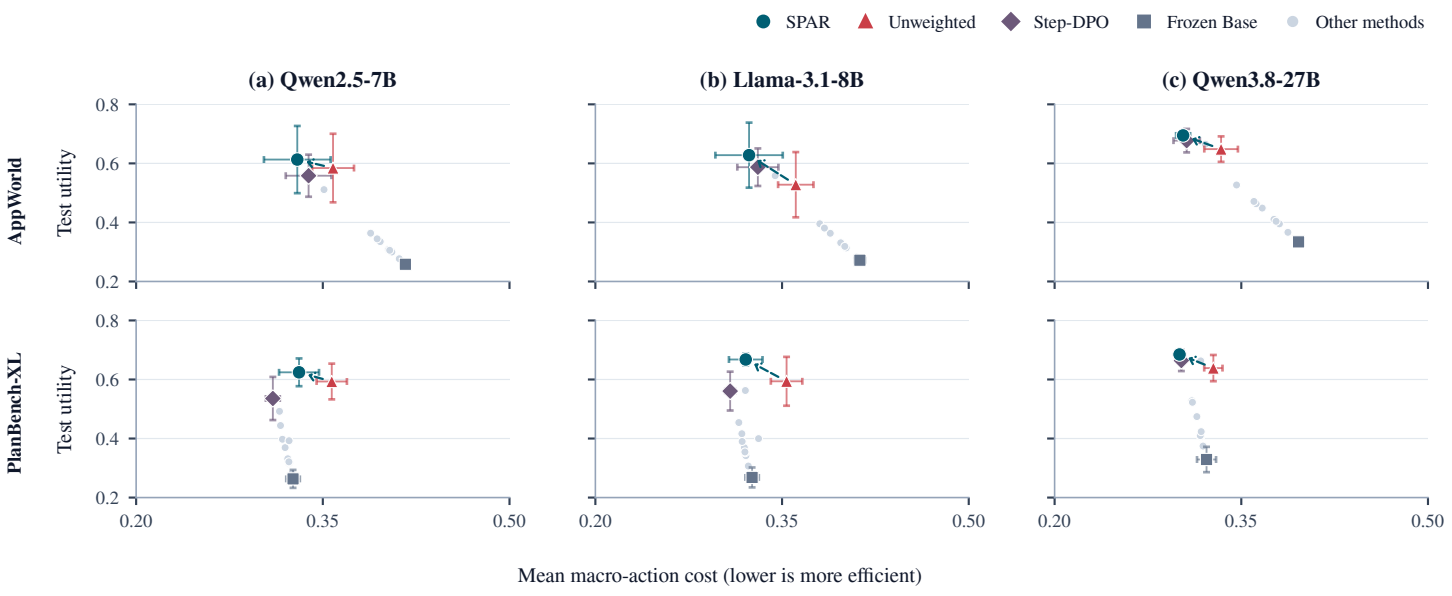
SPAR
Unweighted
Step-DPO
Frozen Base
Other methods
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld
PlanBench-XL
Test utility
0.8
0.6
0.4
0.2
0.20
0.35
0.50
Mean macro-action cost (lower is more efficient)